

Dhruv Nirmal

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PROFESSIONAL SUMMARY

I am a data analyst with a strong foundation in applied statistics, machine learning, and sensitive-data governance, trained through a Master of Data Science at Monash University (Statistics I/II, Applied Forecasting, High Dimensional Analysis, Machine Learning) and sharpened across several years of applied analytical work. I am at my best on ambiguous, data-rich problems: framing the right question, building a rigorous model, and explaining what the results mean to people who did not run the analysis. My statistical work spans multivariable regression, binary classification, time-series forecasting, and anomaly detection, all built and validated on real datasets with real accountability. I am Python-native and actively building R fluency; my statistical foundations transfer cleanly across both languages. I am drawn to the University of Pittsburgh Department of Neurology real-world evidence work because it pairs the kind of careful, consequential analysis I find most rewarding with a training pathway in EHR methods, clinical trial methodology, and applied AI that aligns with where I am heading.

KEY SKILLS

Statistics and Analytics: Multivariable Regression, Logistic Regression, Hypothesis Testing, Statistical Modelling, Anomaly Detection, Time-Series Forecasting, Cross-Validation, Feature Engineering

Programming: Python (scikit-learn, Prophet, Pandas, NumPy), SQL, R (actively developing), MATLAB

ML and Data Science: Classification Models, Class-Imbalance Handling, Hyperparameter Tuning, TF-IDF/NLP, Sentiment Analysis

Data and Governance: Sensitive Data Handling, Data Quality and Validation, ETL Pipelines, Data Cleaning, Large Dataset Management

Visualisation and Communication: Tableau, Power BI, Figures and Tables, Data Storytelling, Non-Technical Stakeholder Reporting

AI and LLM: Claude, ChatGPT, Agentic AI Workflows, Multi-Agent Orchestration, AI Output Evaluation

EXPERIENCE

Data Analyst, Purchasing Index Data Analytics (Comprara Group)

Jun 2024 to Present | Melbourne, Australia

A data and analytics consulting firm serving enterprise clients across Australia and New Zealand.

Run analytical projects end-to-end for five enterprise client accounts: scoping loosely defined problems, gathering and validating large datasets, building statistical and ML models, and translating findings into plain language for senior stakeholders (CFOs, CPOs, category managers). Manage delivery timelines and a regular meeting cadence across all five accounts.

- Designed and deployed a binary classification model (Logistic Regression, one-vs-all; TF-IDF feature engineering, class-imbalance handling, cross-validation, hyperparameter tuning) over approximately A\$12B of client spend across 5 years, cutting a categorisation cycle that previously took 1 to 1.5 months down to a single day run; documented as a repeatable pipeline now being rolled out across clients.
- Built a Prophet time-series forecasting model (Python) incorporating external price drivers (sea-freight trends) to predict 3-month raw-material and chemical inventory requirements for a fresh-produce client, achieving a 12.5 to 14% error margin within the stated tolerance.
- Performed anomaly detection on large procurement datasets, identifying cost-leakage patterns contributing to approximately 15% in client savings; framed findings from ambiguous signals into structured, actionable recommendations communicated to non-technical stakeholders.

- Owned the end-to-end analytics engagement for a restaurant chain (AU/NZ) as sole analyst: structured an ambiguous cost problem into testable hypotheses, ran pricing-movement and cost-leakage analysis, and surfaced approximately A\$2M per year (~30%) in savings in their largest category; presented results directly to CFO, CPO, and general managers in plain language.
- Govern data quality and pipeline integrity for the largest client account (5 global regions, 13 sub-regions, approximately 6M invoice rows refreshed weekly): standardised submission conventions, automated row-count and column-consistency validation, and routed discrepancies back to client and internal teams, maintaining strict accountability for data accuracy throughout the pipeline.
- Built and maintained 20+ automated ETL pipelines (Python, SQL, Azure) and 16+ Tableau dashboards for enterprise clients; developed agentic AI workflows (Claude) for report generation, sentiment analysis, and multi-agent dataset QA with human-scrutiny gates at each sign-off step.

Data Engineer Intern, Victorian Centre for Data Insights (VCDI)

Aug 2023 to Nov 2023 | Melbourne, Australia

- Built a distributed anomaly-detection pipeline (Databricks, PySpark) improving detection accuracy 20%; engineered scalable transformation layers for cloud procurement data.
- Deployed a Power BI analytics solution adopted by senior Department of Transport stakeholders; presented model architecture and outcomes to cross-functional non-technical audiences.

Research Data Analyst, Terminal Ballistics Research Laboratory (TBRL), DRDO

Jan 2021 to Jul 2021

- Analysed blast-wave sensor data using statistical models (MATLAB), improving noise-level prediction accuracy 20% and contributing to a 10% improvement in defence equipment design efficiency.
- Applied signal filtering (Butterworth filter) and statistical modelling to noisy sensor data; compiled findings into a 70-page structured report translating quantitative results into clear recommendations for engineering teams.

PROJECTS

ML Spend-Categorisation: Logistic Regression Classifier (Python, scikit-learn, TF-IDF)

- Built a one-vs-all binary classifier with feature engineering (text, spend value, line count), class-imbalance handling, cross-validation, and hyperparameter tuning over approximately A\$12B of client spend data; documented as a repeatable pipeline now being adopted across the client base.

Inventory Demand Forecasting: Prophet Time-Series Model (Python, Prophet, Dashboard)

- Developed a 3-month demand forecast for raw materials and chemicals incorporating sea-freight price trends as external drivers; achieved a 12.5 to 14% error margin within the stated tolerance, delivered on a BI dashboard.

MyFacit: Analytics and Decision-Support Platform

- Designed and built a centralised analytics platform for a hospitality business integrating 8+ data sources; AI-driven workflows for data processing, reporting, and short-term operational forecasting; in active pilot use with weekly feedback cycles assessing real-world performance.

EDUCATION

Master of Data Science, Monash University | Feb 2022 to Dec 2023

Coursework: Statistics I/II, Machine Learning, Applied Forecasting, High Dimensional Analysis, Communicating with Data

Bachelor of Engineering, Thapar University | May 2017 to May 2021

Coursework: Statistics, Mathematics I/II/III, Analog and Digital Electronics